

Experiment 13

13.1.1 Fibonacci Sequence using Modules

Algorithm :

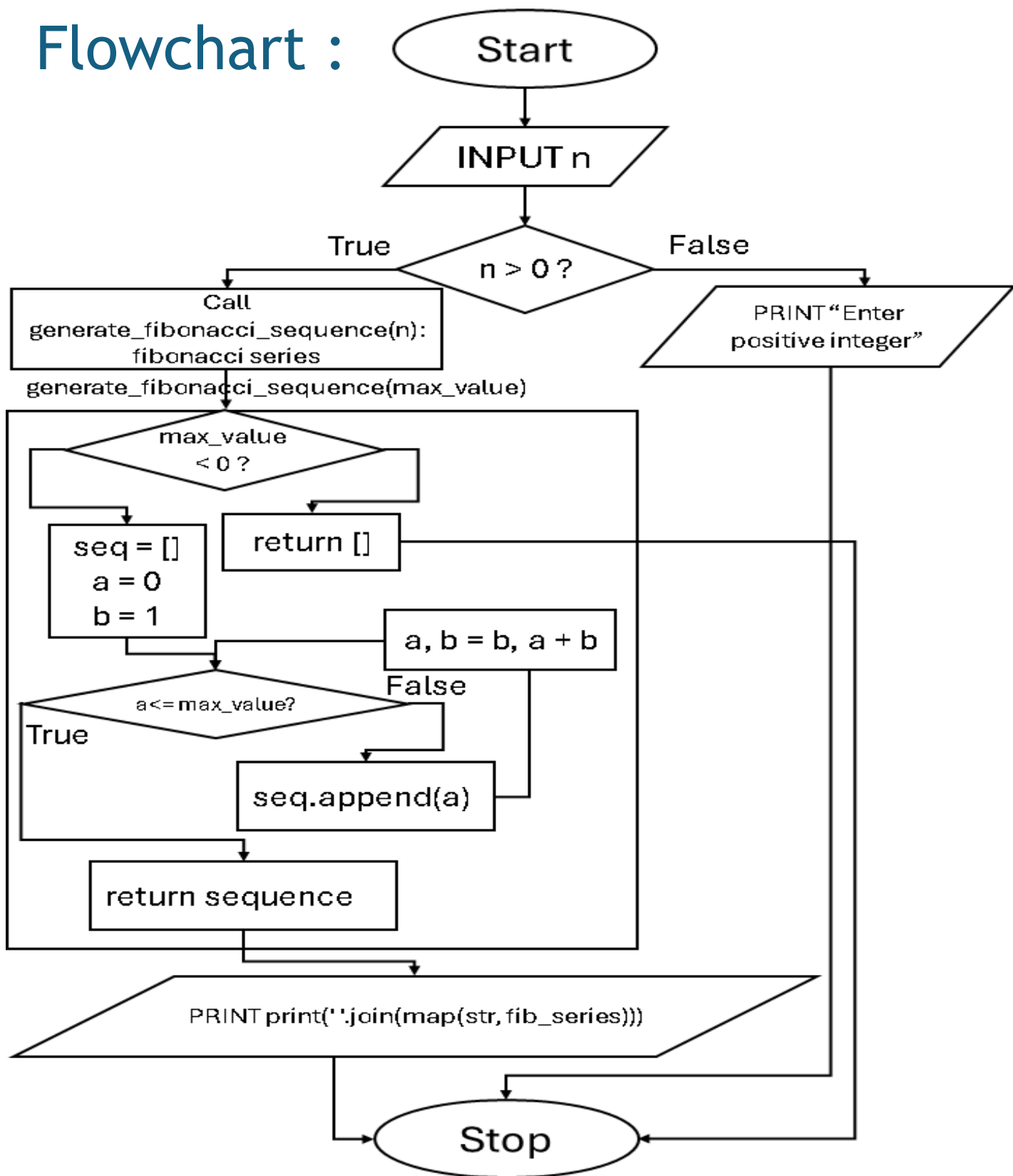
Main Program

1. Start
2. Input integer n
3. Check if $n > 0$
 1. If **Yes**, go to Step 4
 2. If **No**, go to Step 7
4. Call function generate_fibonacci_sequence(n)
5. Store result in fib_series
6. Print elements of fib_series separated by space
7. Display message: "Please enter a positive integer" (if $n \leq 0$)
8. Stop

generate_fibonacci_sequence(max_value)

1. Start
2. If $\text{max_value} < 0$
3. Return empty list []
4. Initialize:
5. $\text{seq} = []$
6. $a = 0, b = 1$
7. Repeat while $a \leq \text{max_value}$
8. Add a to seq
6. Update values:
 1. $a = b$
 2. $b = a + b$
10. Return seq
11. Stop

Flowchart :



Code :

main.py

```
import fibonacci_module
n = int(input())
```

```
if n > 0:
    fib_series =
    fibonacci_module.generate_fibonacci_sequence(
n) print(' '.join(map(str, fib_series)))
else:
    print("Please enter a positive integer")
```

fibonacci_module.py

```
def
generate_fibonacci_sequence(max_value
): if max_value < 0:
    return []
seq = []
a = 0
b = 1
while a <= max_value:
    seq.append(a)
    a, b = b, a + b
return seq
```

Execution :

CODETANTRA

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Programming and Problem Solving Lab - TE7287 - II Sem - 2025

13.1.1. Fibonacci Sequence using Modules

Write a Python program using modular programming to generate the Fibonacci sequence up to a given maximum value.

Module Requirements

Create a module `fibonacci_module.py` with a function `generate_fibonacci_sequence(max_value)` that returns list of all Fibonacci numbers less than or equal to the maximum value.

Main Program Requirements

Create `main.py` that performs the following:

Import the `fibonacci_module`.

Read an integer `n` from user input.

Validate if `n` is positive.

If valid, call the module function and display the Fibonacci sequence.

If invalid, display an error message.

Input Format:

Single integer `n` representing the maximum value for Fibonacci sequence.

Output Format:

If `n > 0`, print the Fibonacci sequence as space-separated numbers.

If `n ≤ 0`, print: "Please enter a positive integer".

Note:

Two files are provided: `fibonacci_module.py` and `main.py`. The `main.py` file handles input and output. You need to complete both files based on the question requirements.

Sample Test Cases

main.py

fibonacci_...

1

2

3

4

5

6

7

8

import fibonacci_module

n = int(input())

if n > 0:

fib_series = fibonacci_module.generate_fibonacci_sequence(n)

print(' '.join(map(str, fib_series)))

else:

print("Please enter a positive integer")

Average time

0.004 s

3.50 ms

Maximum time

0.005 s

5.00 ms

4 out of 4 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1

Expected output

0 1 1 2 3 5 8

Actual output

0 1 1 2 3 5 8

Test case 2

Test case 3

Terminal

Test cases